

Sudip Sharma

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Education

- Morgan State University**, *Master of Science in Advanced Computing* Dec 2026 | Baltimore, US
Coursework: Design And Analysis Of Algorithms, Software Engineering, Cloud Computing
- Kathmandu University**, *Bachelor of Science in Computer Science* 2018 – 2023 | Dhulikhel, Nepal
Coursework: Data Structures and Algorithms, Algorithms and Complexity, Database Management Systems

Professional Experience

- Morgan State University**, *Research Assistant* Sep 2024 – May 2025 | Baltimore, US
• Built interpretable time-series predictor on NIH Chest X-ray, measured by AUC 0.89 / PR-AUC 0.31, by a CNN-Transformer pipeline with SHAP/Grad-CAM.
- EB Pearls**, *Software Engineer* Jul 2023 – Aug 2024 | Lalitpur, Nepal
• Implemented login screens, alerts, and navigation while integrating an OTP/token system, resulting in significantly improved user experience and enhanced security in **SwiftUI**
• Integrated Google API for calendar events fetching, achieving a 63% increase in user experience in SwiftUI
• Utilized **NestJS** framework to build **GraphQL** APIs for both mobile and admin side projects, improving data management and performance by 37%
- EB Pearls**, *Software Engineering Intern* Apr 2023 – Jul 2023 | Lalitpur, Nepal
• Learned and implemented **Swift** for iOS application development, utilizing both SwiftUI and **UIKit** frameworks to create engaging and user-friendly interfaces
• Acquired a strong foundation in **JavaScript** fundamentals, applying the knowledge to implemented backend solutions using **Node.js** and **Express**
• Recognized by the manager for fast learning and consistent research into new technologies to integrate into projects for improved performance
- Center for Equitable Artificial Intelligence and Machine Learning Systems**, *Graduate Mentor (Morgan State University)* May 2025 – Aug 2025 | Baltimore, US
• Led a multimodal deep learning project (1D-CNN, Transformer, 2D-CNN) to classify cardiovascular conditions from ECG signals with over 90% model accuracy
• Guided undergraduates in feature extraction (time, frequency, time-frequency) and model fusion, improving ensemble performance and interpretability
• Presented model outputs using explainability tools (GRAD-CAM) and optimized preprocessing for enhanced signal quality and fairness

Skills

- Swift**
Frameworks: Swift UI, UIKit
- Javascript/Typescript (Node.js)**
Frameworks: NestJS, Express (GraphQL/REST API)
- Python**
Libraries: Matplot.lib, Pandas, Sklearn, Tensorflow
- Cloud Computing**
Amazon Web Service: S3, EC2, SageMaker, VPC

Projects & Leadership

- 2025 National Symposium on Equitable AI**, Baltimore, US
Research Lead & Presenter, Morgan State University
• Developed fairness-aware CNN for skin lesion classification using the HAM10000 dataset
• Improved recall for melanoma (29%→54%) and reduced bias across skin tones using Fairlearn metrics
• Applied adversarial debiasing, class weighting, and Grad-CAM validation
- Mastercard Center for Inclusive Growth**, *AUC Data Science Initiative* Atlanta, US
• Earned 3rd place in the 2024 Mastercard Center for Inclusive Growth x AUC Data Science Initiative
• Led a project analyzing disparities and proposing strategies for economic inclusion using data science
- IT MEET 2022**, *Non-tech Events Lead* Dhulikhel, Nepal
• Led events like a Career Fair, Alumni Meet, Blood Donation, Acoustic Night, and Idea Pitching, fostering community engagement and professional networking, resulting in increased participation by 21%